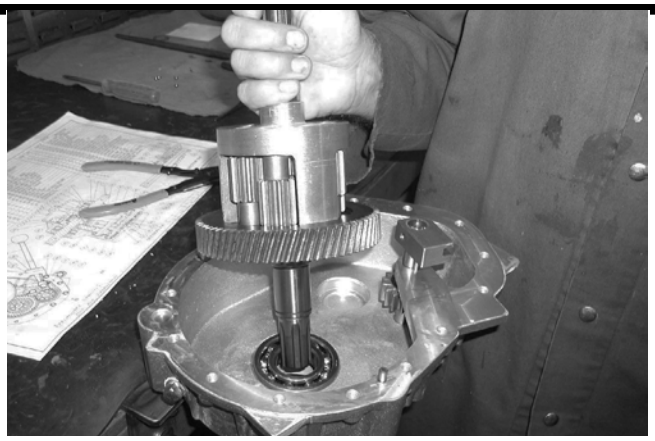
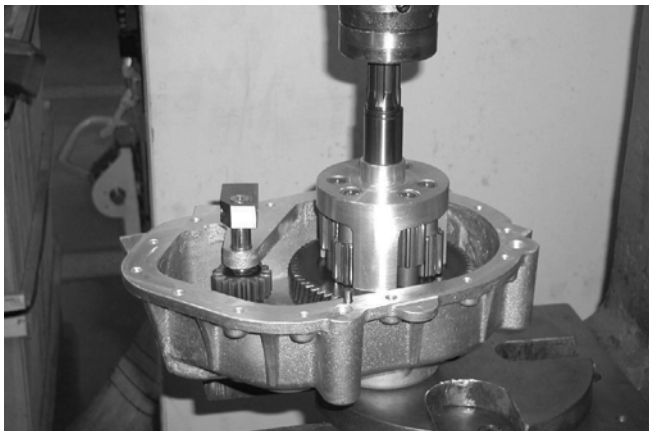
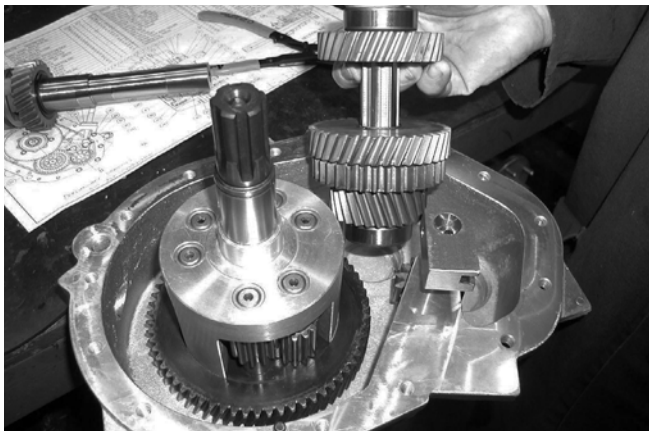
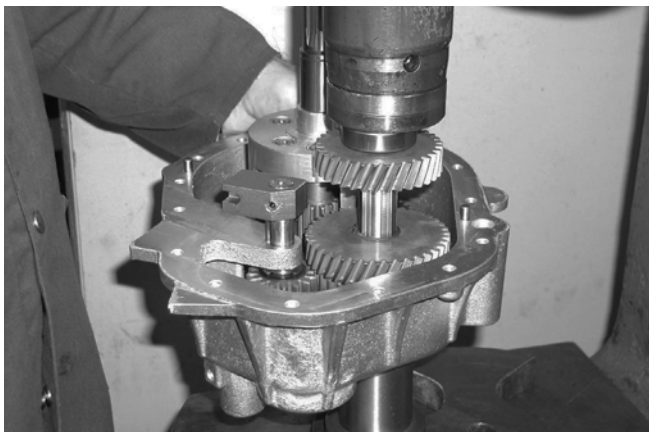


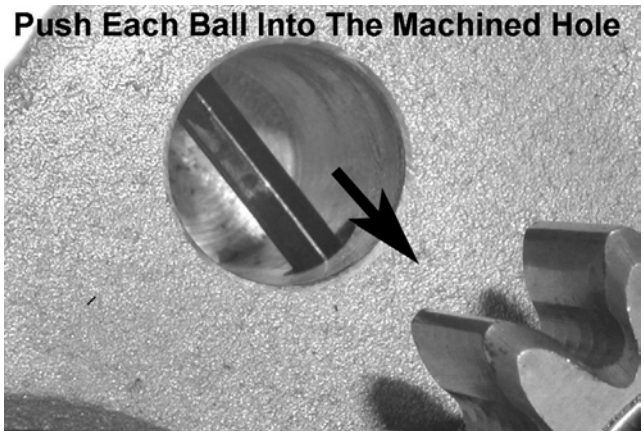




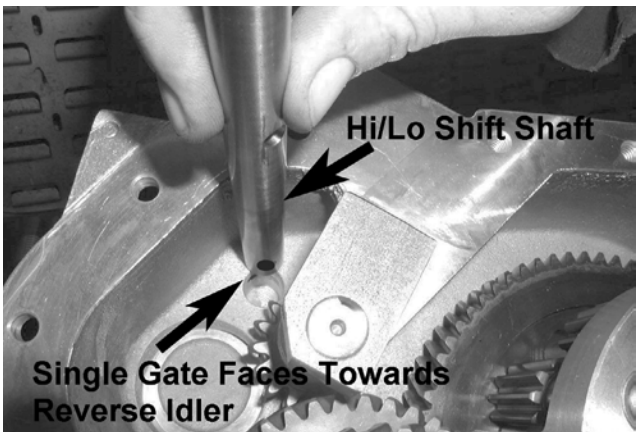


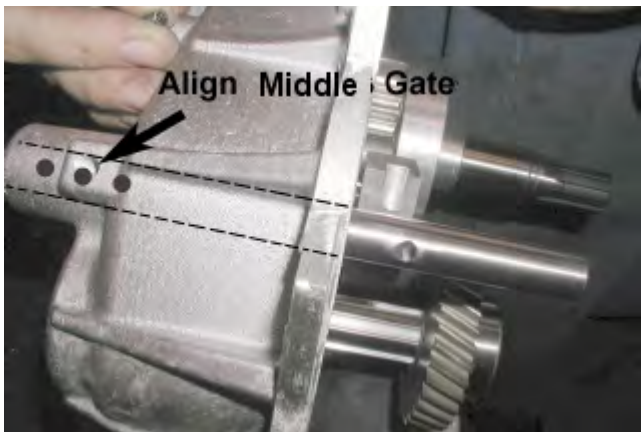


18. Next, push the Reverse Shift Shaft and Block down against the Housing boss (out of reverse). Install (3) 109-05 Steel Balls into the Hi/Low Shift Shaft cavity of the Housing, and push each one into the machined hole at the Reverse Shift Shaft. These (3) Steel balls will be located between the Reverse Shift Shaft and Hi/Low Shift Shaft.

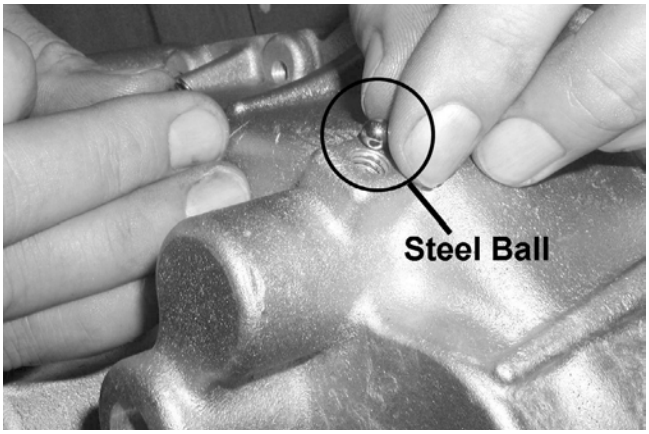


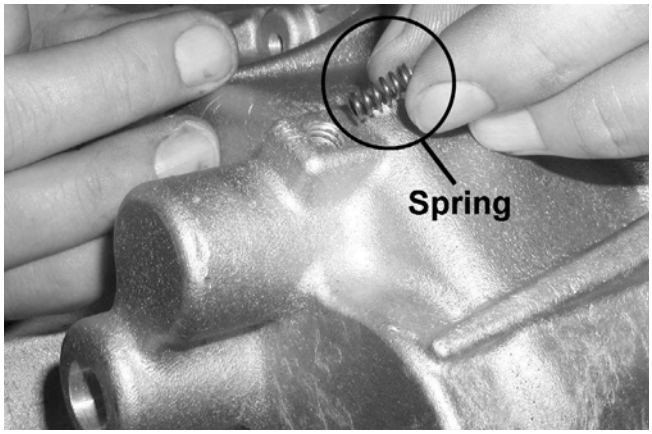
19. Install the 34-113 Hi/Low Shift Shaft into the Housing with the 3 machined gates facing to the outside of the housing and the 1 machined gate facing the Reverse Shift Shaft.



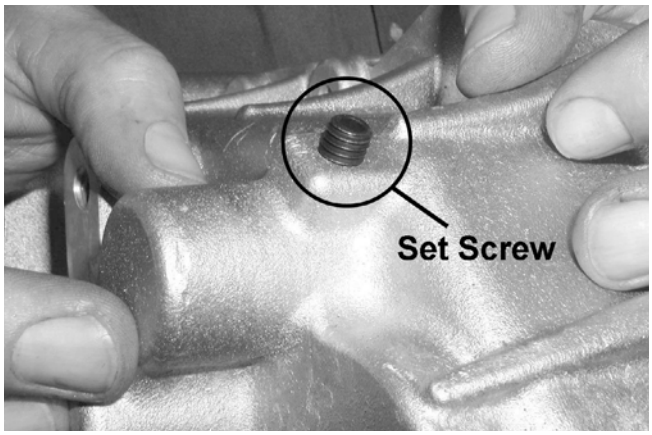


20. Align the middle gate in the Hi/Lo Shift Shaft with the access hole in the casting and install a 109-05 Steel Ball and 34-32 Spring.



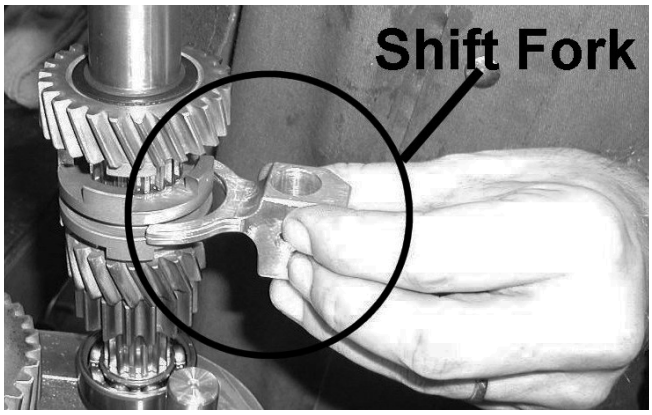


21. Apply Blue LOCTITE to the threads of the 112-145 Set Screw and tighten into the Housing approximately 2 1/2 turns. This may need to be adjusted later depending on how hard or easy the transmission shifts when tested.

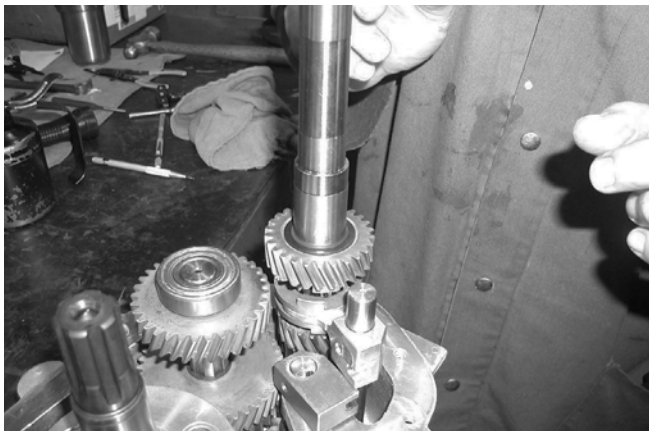


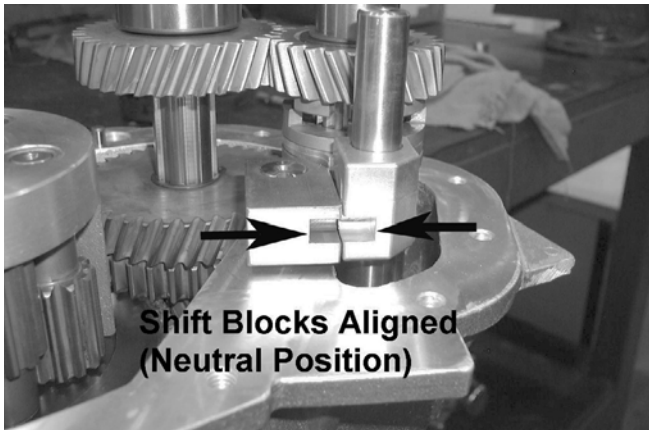
22. Assemble the 34-116 Shift Fork to the Dog Clutch on the Input Shaft Assembly.

23. Raise the Reverse Shift Block up (into reverse). Align the the machined hole of the Hi/Lo Shift Block Fork, with that of the Hi/Lo Shift Shaft, and install the complete Input Shaft assembly into the housing.



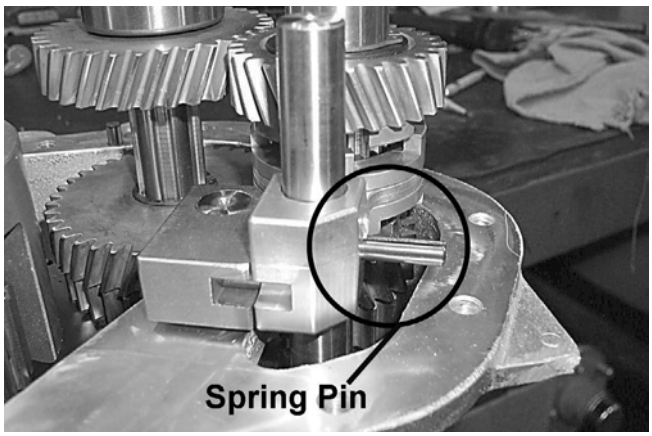


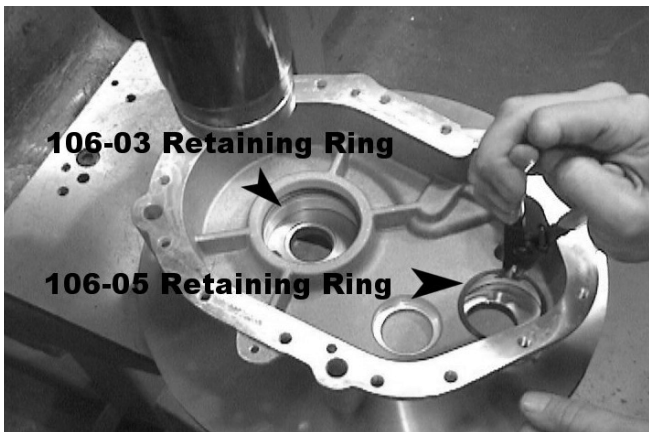




24. Place the transmission into the neutral position.

25. Secure the Hi/Lo Shift Fork to the Shift Shaft using a 104-39 Spring Pin.





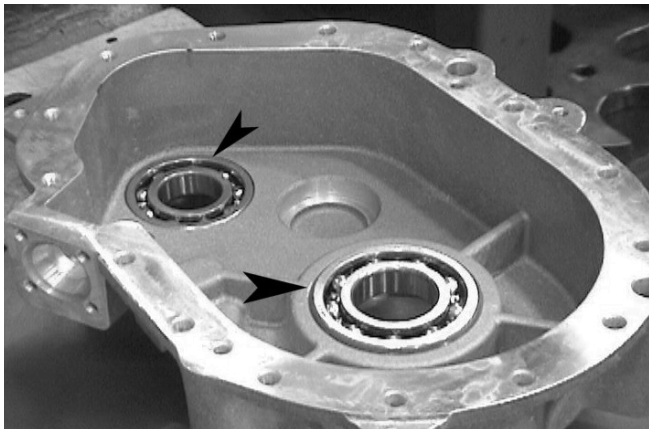
26. Install a 106-03 Retaining Ring to the top groove of the 34-101 Left Transmission Housing at the output shaft opening, and a 106-05 Retaining ring at the input shaft opening

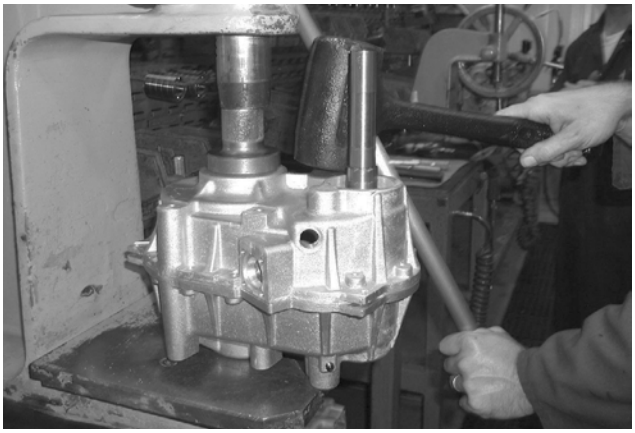
27. Place a 114-01 shim to the top of the 106-03 Retaining Ring.

28. Press a 101-03 & 101-16 Bearing into the Left Transmission Housing.

29. Apply Black 598 LOCTITE RTV Silicon Gasket Maker around the perimeter of the Transmission Housing

30. Press the Left Hand Housing to the Right Hand Housing at the arbor press. Tap with a soft face mallet while pressing at the output shaft area. This will ensure that the housing is pressed on squarely and evenly to the lower half.

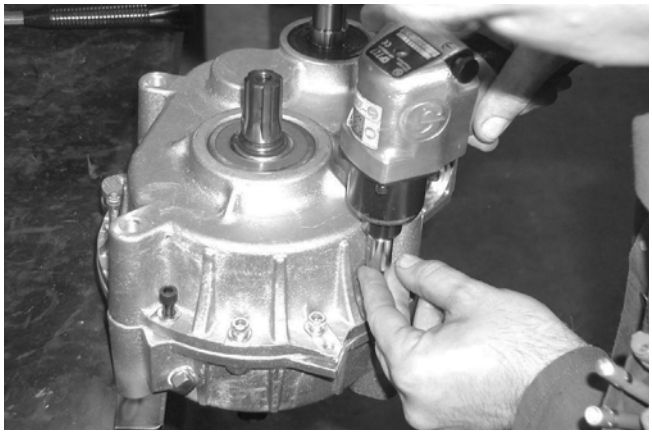


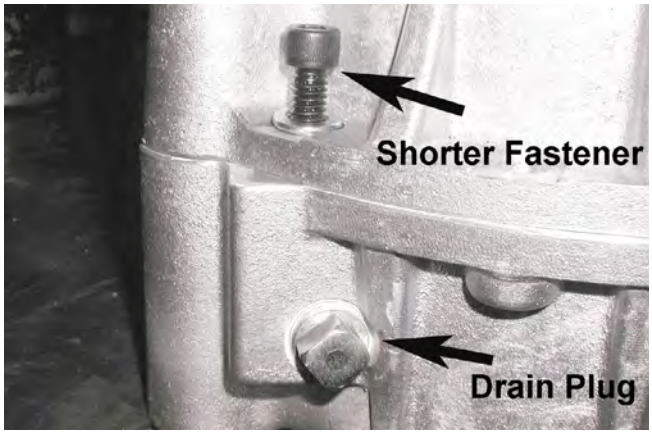


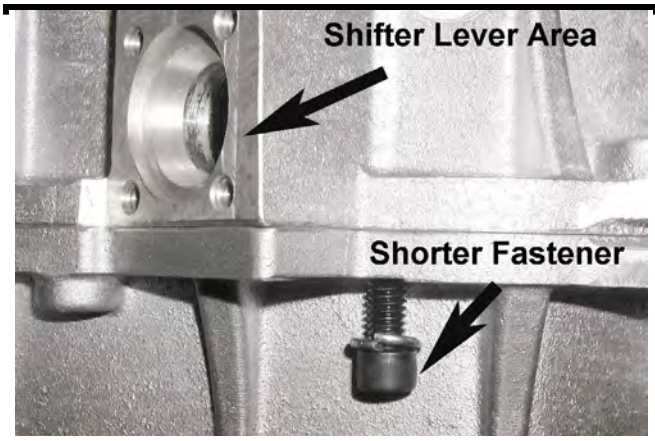
31. Secure the halves of the transmission together using (10) 112-143, (1) 112-144 fasteners & (11) 108-05 lockwashers installed from the left side of the Housing. Three (3) 112-143, (1) 112-144 fasteners and (4) 108-05 lockwashers are installed from the right side of the Housing. **Torque to 20FT-LBS**

NOTE

Two of these fasteners are shorter and are assembled into the transmission casting at blind holes. One is assembled at the drain plug area from the left side of the housing and the other at the Shifter Lever area from the right side of the housing.





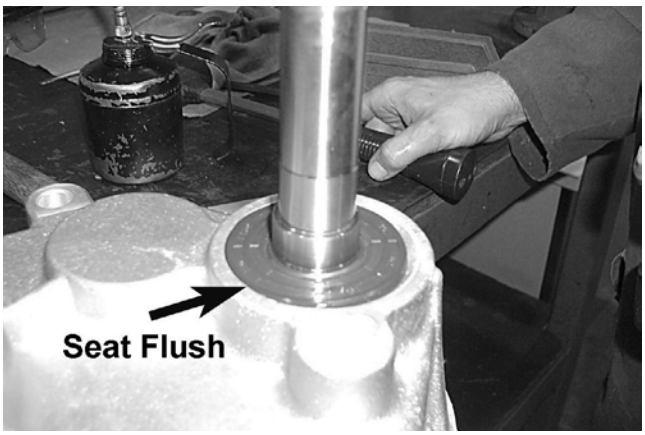


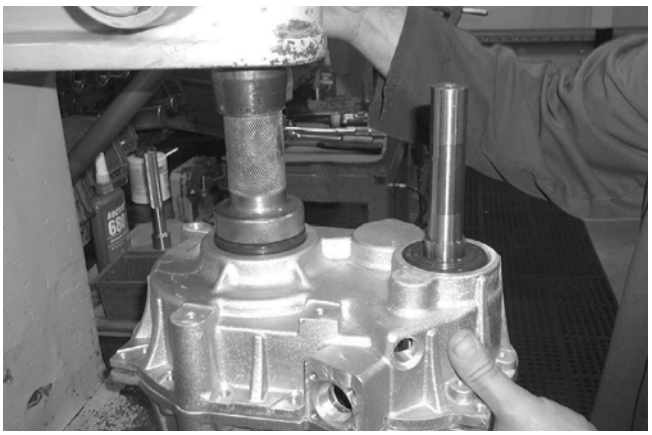


32. Install the 102-05 Oil Seal to the Input Shaft.

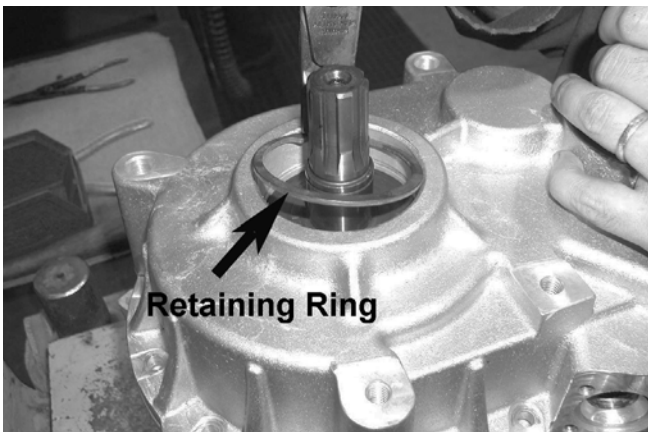
33. Install the 102-03 Oil Seal to the Output Shaft. Seat below the Retaining Ring groove.

34. Seat a 106-03 Retaining ring to the machined groove just above the previously installed Oil Seal.

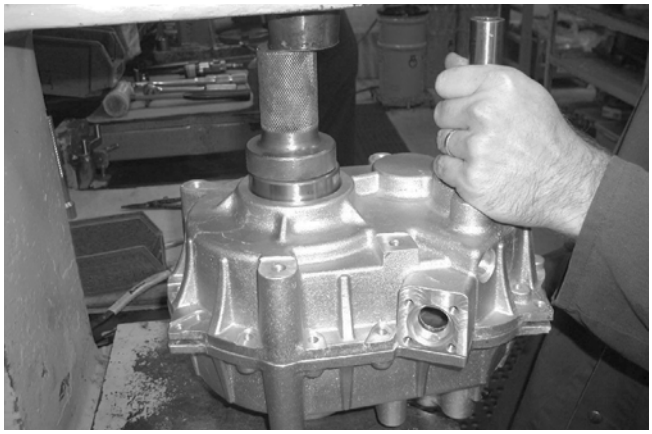


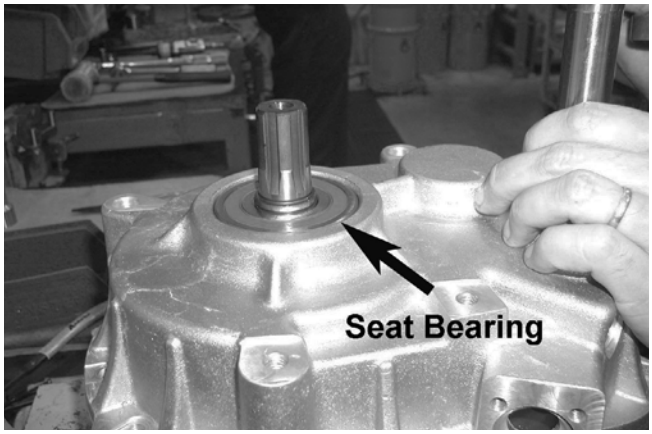






35. Install a 101-05 Bearing to the Output Shaft.





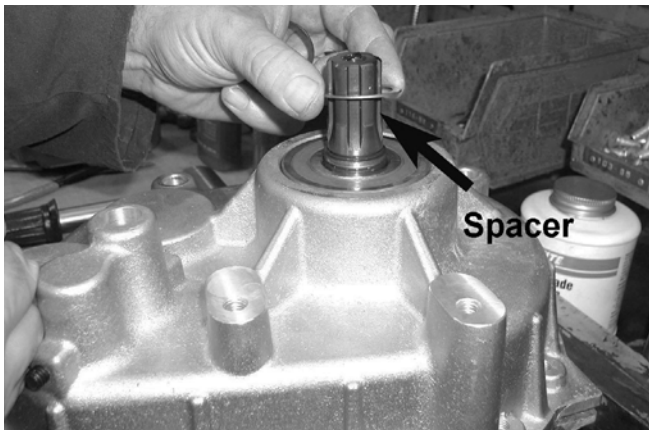
36. Repeat Steps 36 through 38 for the other Output Shaft.

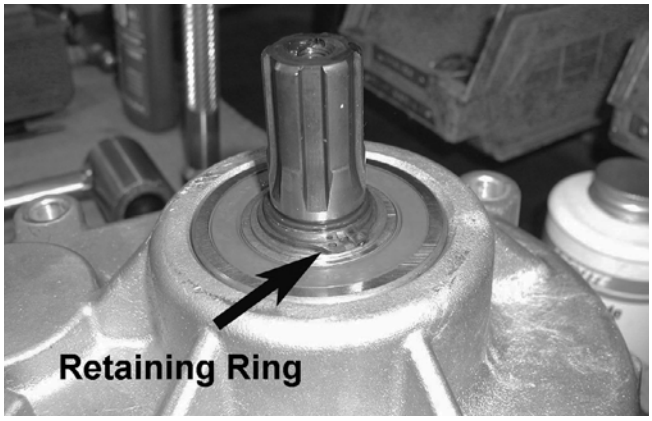
37. Using the Output Shaft puller, pull each of the Output Shafts. Install a 114-04 Shim under the 107-02 Retaining Ring at the Output Shaft on the Left Side Housing. Install a 114-05 Shim under the 107-02 Retaining Ring at the Output Shaft on the Right Hand Side Housing.

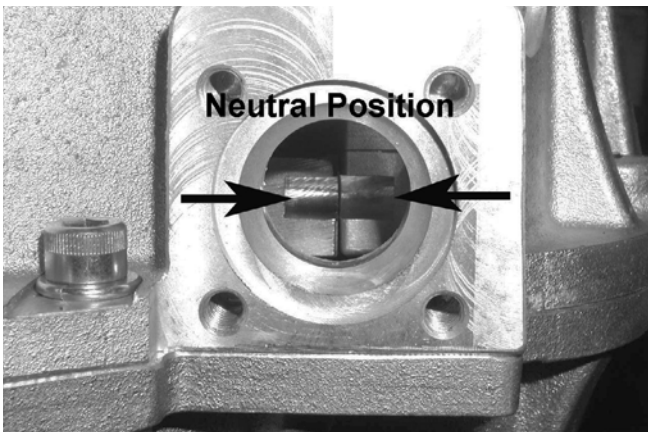
IMPORTANT

Ensure that all retaining rings are seated properly





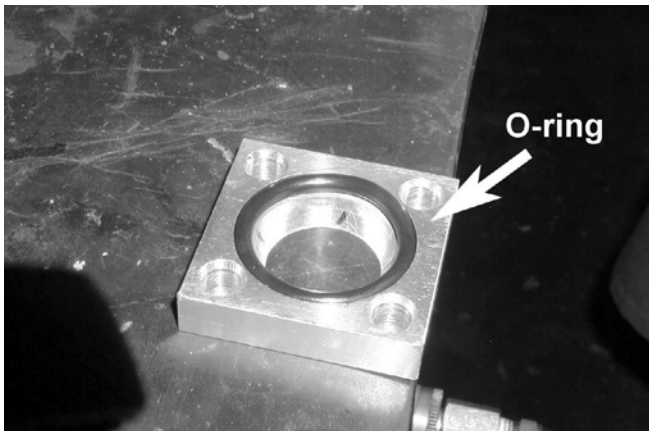




38. Locate the shift lever opening and place the transmission shift blocks in the neutral position.

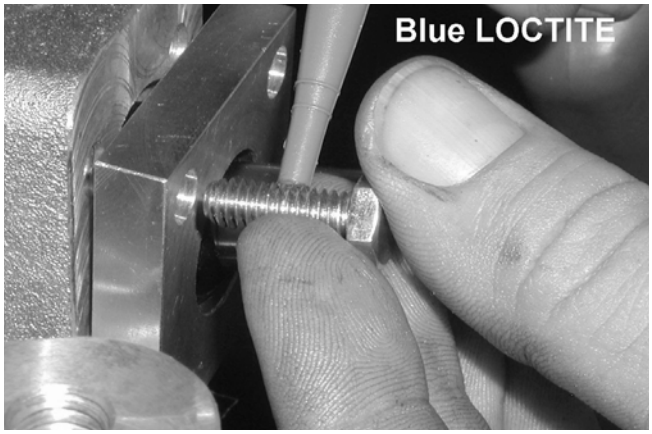
39. Place the 113-11 O-ring into the 34-118 Shifter Plate.

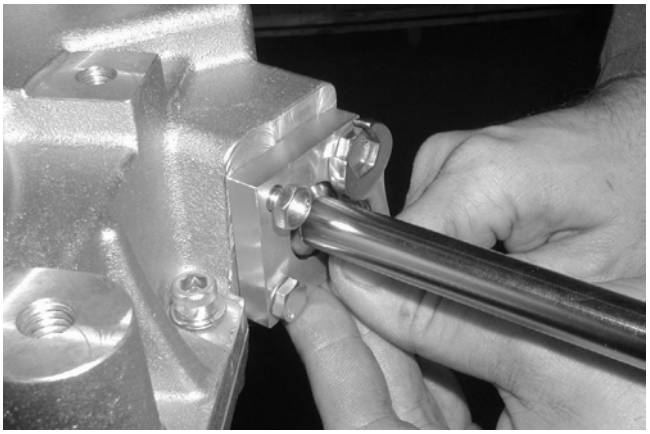
40. Assemble the 34-117 Shift Lever to the Shifter Plate. Apply a small amount of axle grease to the ball area of the shifter lever.





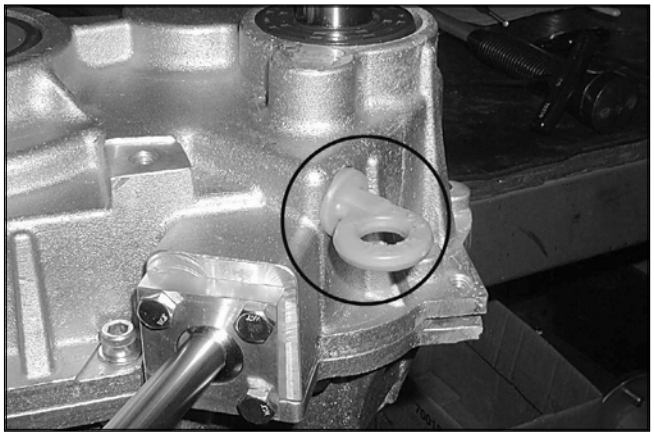
41. Apply Blue LOCTITE to each of the 112-04 Fasteners and secure the Shift Lever assembly to transmission.





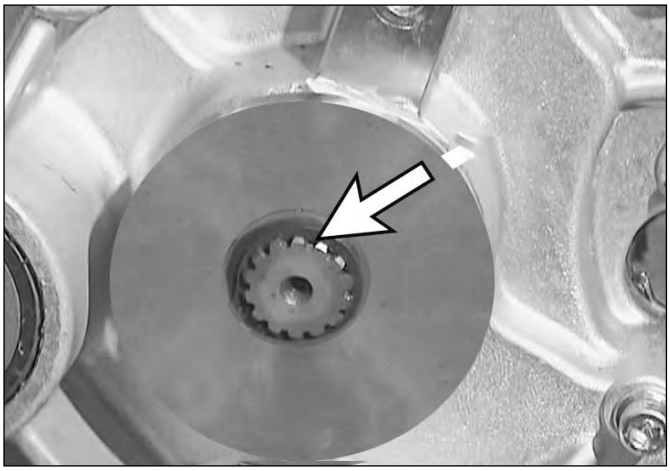
42. Install the CE-035 Dipstick.

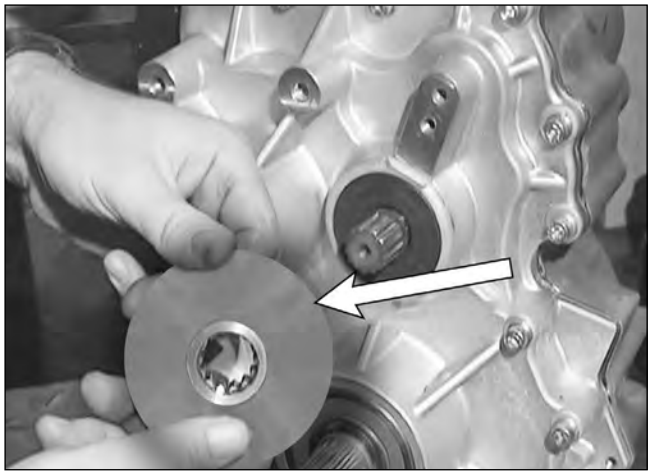
43. Fill the transmission with 1.1 L (38.7 oz of 80 W 90 Gear Lube HYPOY-C or ARGO brand 75W90 Synthetic lube (Part No.130-104).



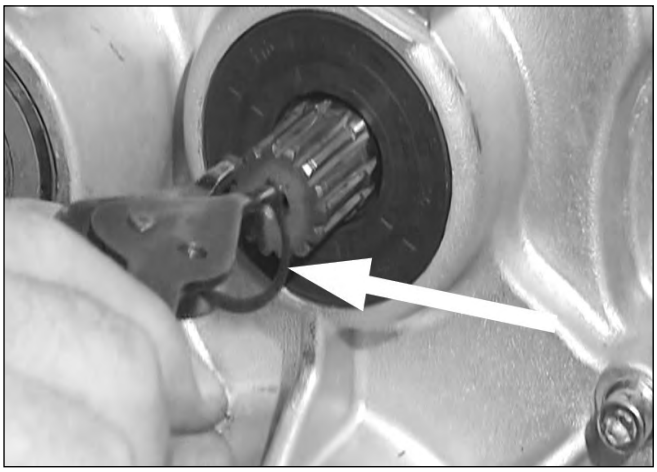
34-200 Transmission Disassembly

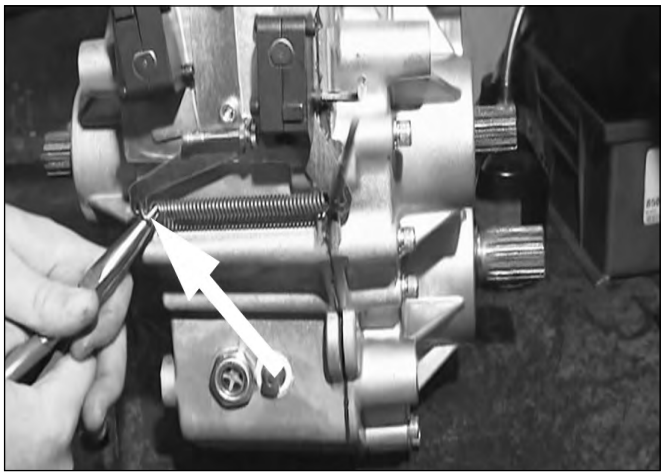
1. Locate the E-brake disc on the transmission. Remove the outer retaining ring.
2. Slip off the e-brake disc Take this opportunity to inspect it. Inspect the drum surface, the spline or any cracks that may be present.



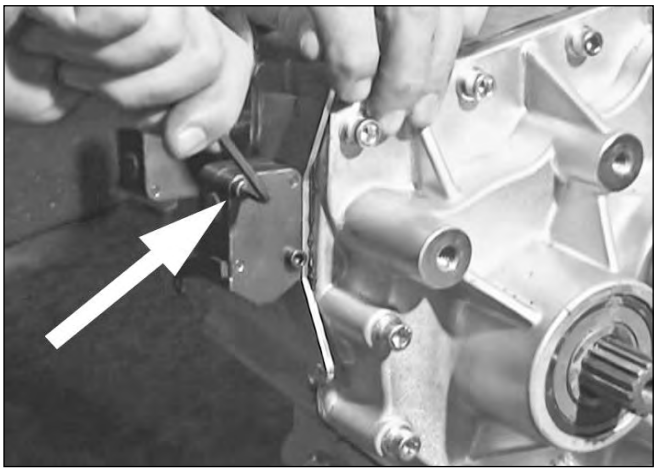


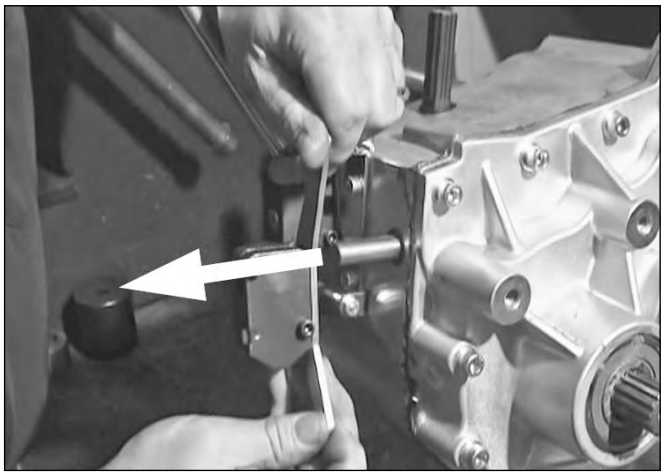
3. Remove the inner Retaining Ring.
4. Disconnect the extension spring from the spring mount bracket.



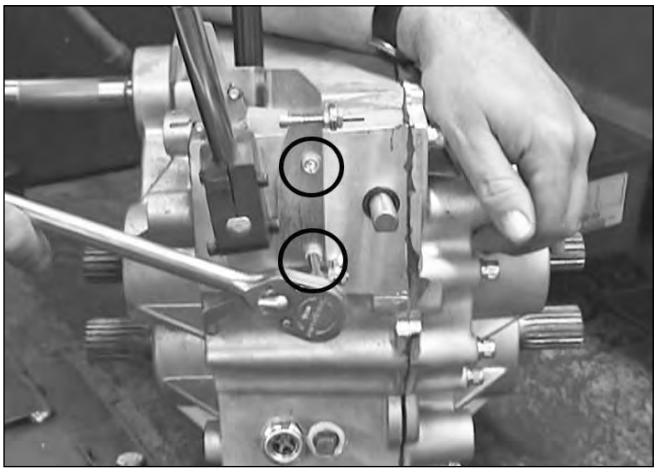


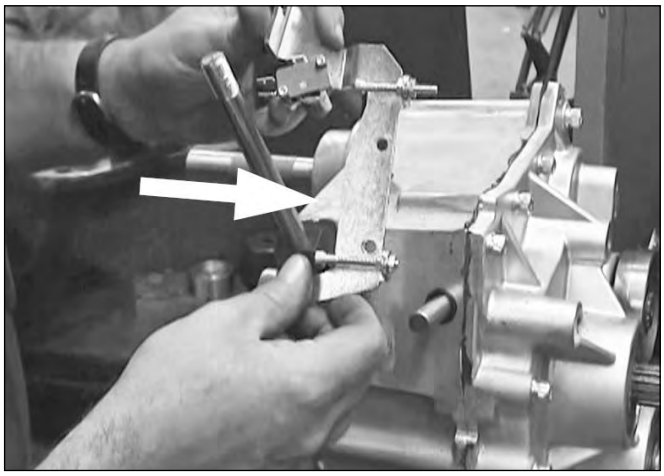
5. Loosen off the fasteners securing the shift lever clamp to the high/lo shift shaft and slide it off of the shift shaft. The spring mount bracket will come off attached to the shift clamp. *See following page.*



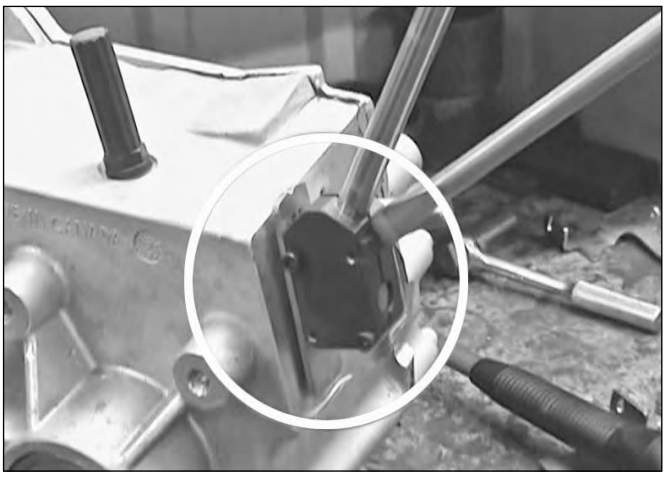


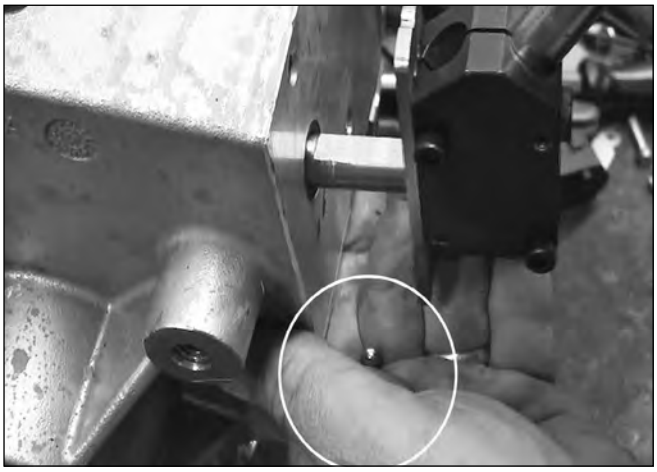
6. Remove the 2 fasteners that are securing the second spring mount bracket to the transmission housing. Set this bracket aside to a clean work area.

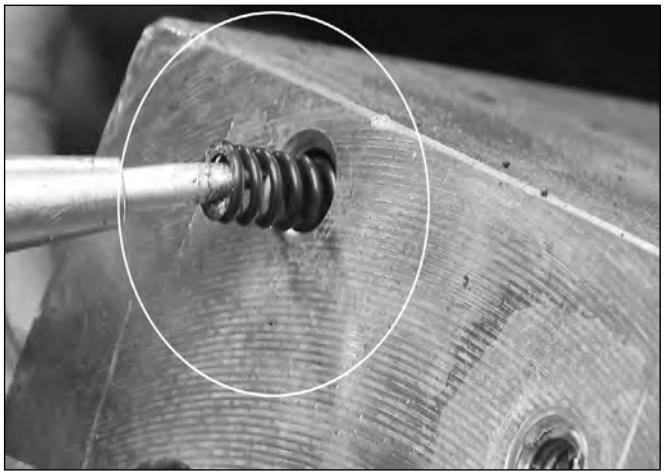




7. Loosen off the 4 fasteners securing the shift lever clamp to the forward reverse shift shaft. Be aware when removing this assembly that there is a small ball and spring located in behind the detent plate. As you pry the shift block with a slot screwdriver you will notice that the assembly will slowly spring forward.





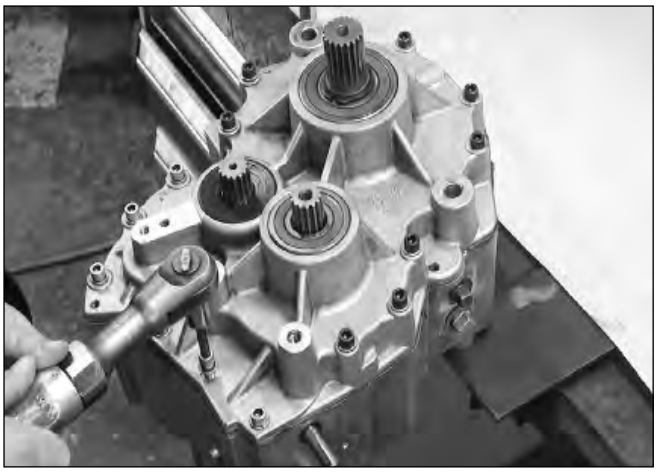


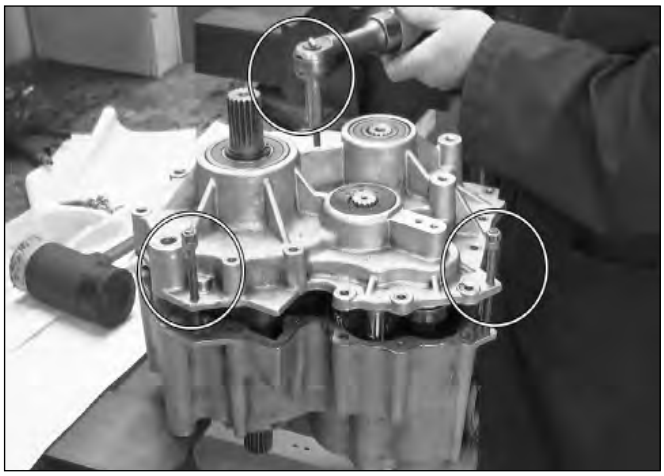
8. On earlier transmissions, as you slide the shifter assembly off of the shaft, be prepared to catch the ball in your hand. (Later transmissions utilize a contained ball and spring insert threaded into the housing). Remove the spring from the transmission housing(earlier transmissions) This ball and spring provides the shift gate notches you feel when you are shifting from forward to neutral to reverse.
9. Place the transmission to the transmission holding fixture and remove all fasteners securing the right hand housing to the left hand side.

⚠ NOTE

Fasteners used in the top half of the transmission are shorter than the ones used in the lower half of the transmission

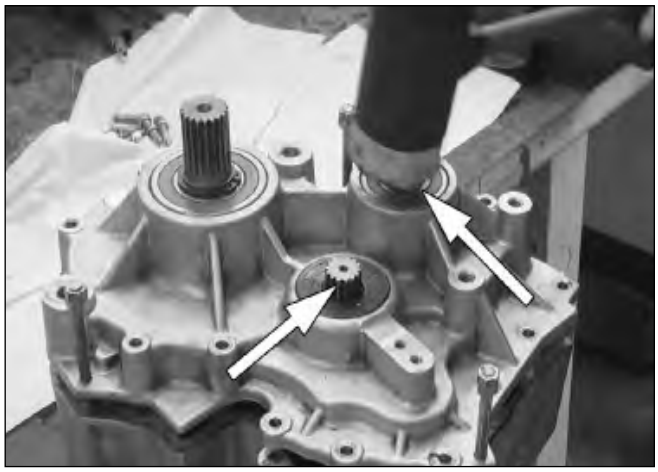
10. To separate the housing there are 3 locations provided on the right hand housing to install some threaded rod. As the threaded rod is turned in at these locations, the cover will begin to separate.

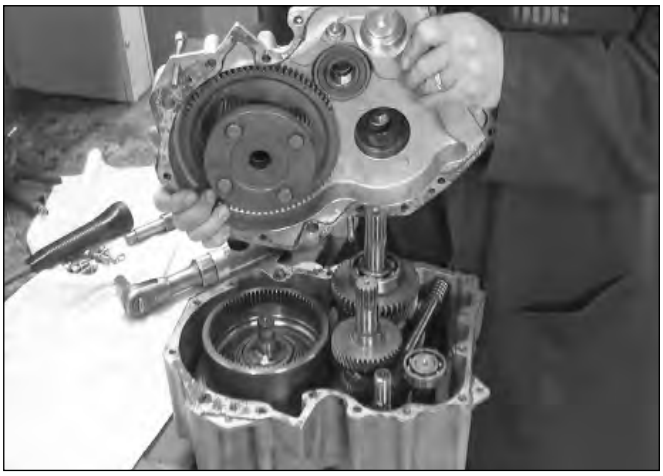




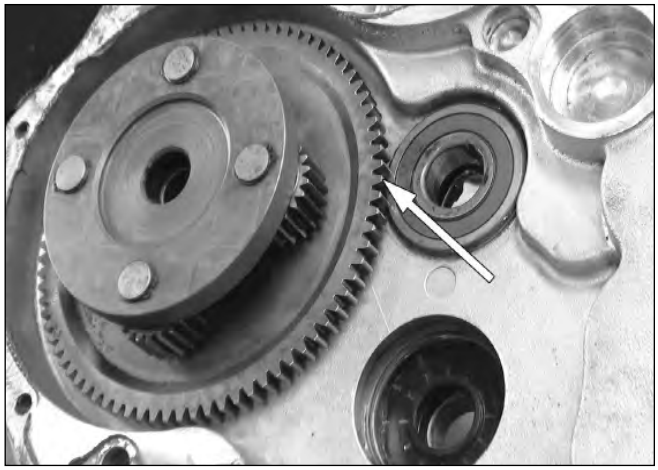
⚠ NOTE

To be sure that the cover separates evenly, thread each of these a little at time alternating between the 3. As the housing begins to separate, periodically tap between the idler shaft and brake shaft.





11. Once the right hand housing is free from the left hand housing,separate and place on the bench. .
12. Remove the threaded rod from each of the three locations on the right hand housing.

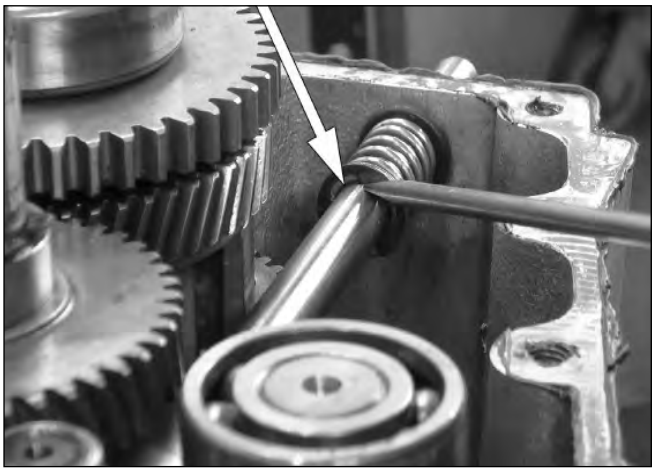


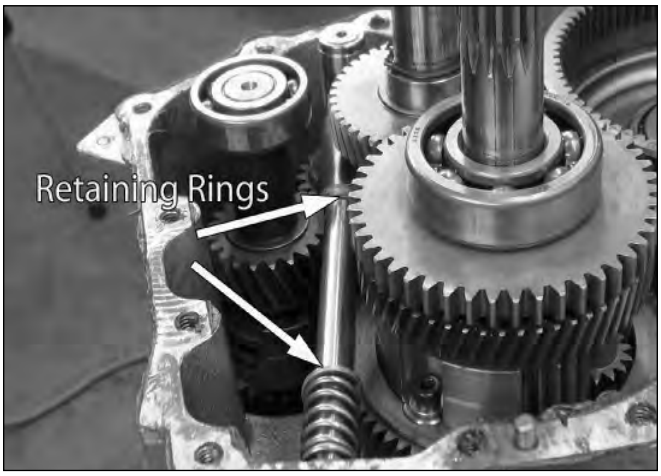


NOTE

Generally the right hand output shaft stays with the right hand housing. The sun gear from that assembly may come into contact with one of the bearings left in the housing. Use a soft face hammer and a bearing seating tool to re-seat that bearing back into the housing.

13. Next locate and remove each of the retaining rings that are securing the hi/lo shift shaft in the housing.





14. Pull the hi/lo shift shaft from the transmission. As the shift shaft is pulled from the transmission,remove both shift forks, spring, washers and spacer as well.

NOTE

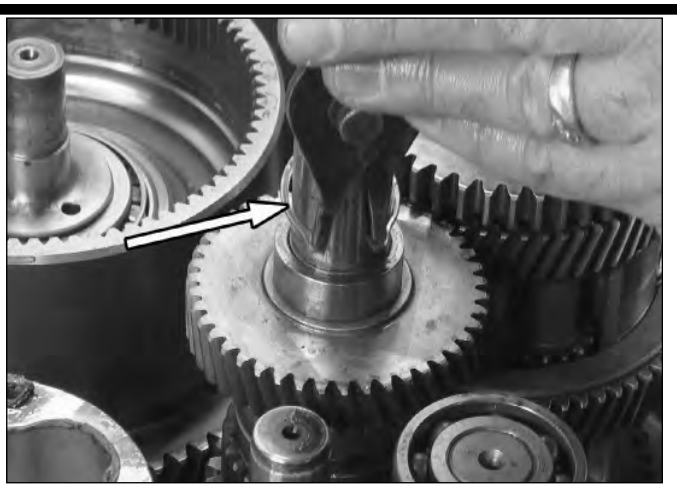
You may find it beneficial to install half of a 34-232 shift fork on to the reverse shift shaft, this will ease in helping to pull out the shift shaft from the housing.



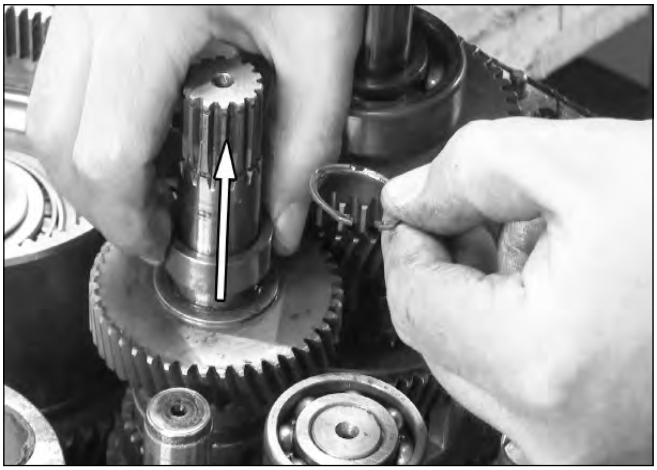


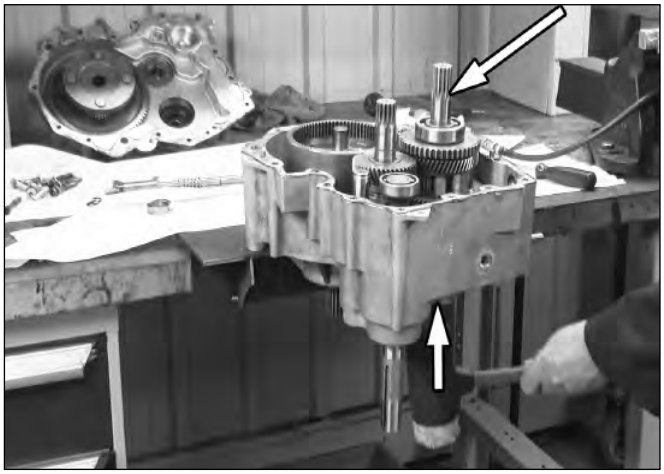
15. As the shift shaft is pulled from the transmission, remove both shift forks, spring, washers and spacer as well. Take the opportunity to reassemble these components back to the shift shaft. *See following page*.





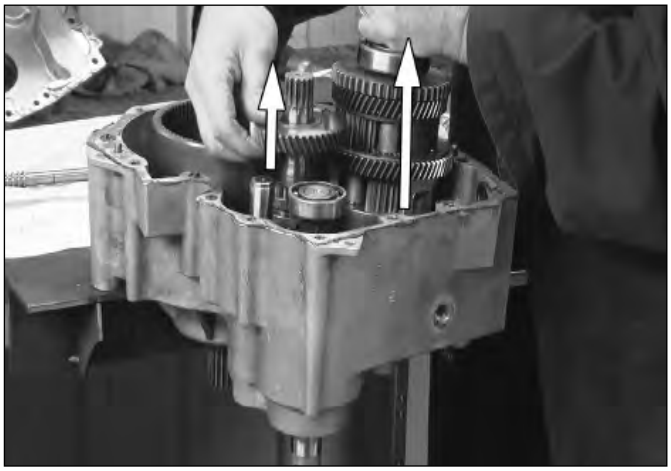
16. Remove the retaining ring securing the spacer on top of the 45 tooth helical pinion of the idler shaft assembly. Remove the spacer as well.
17. From below the transmission tap on the steering brake differential shaft to remove it from the housing.





NOTE

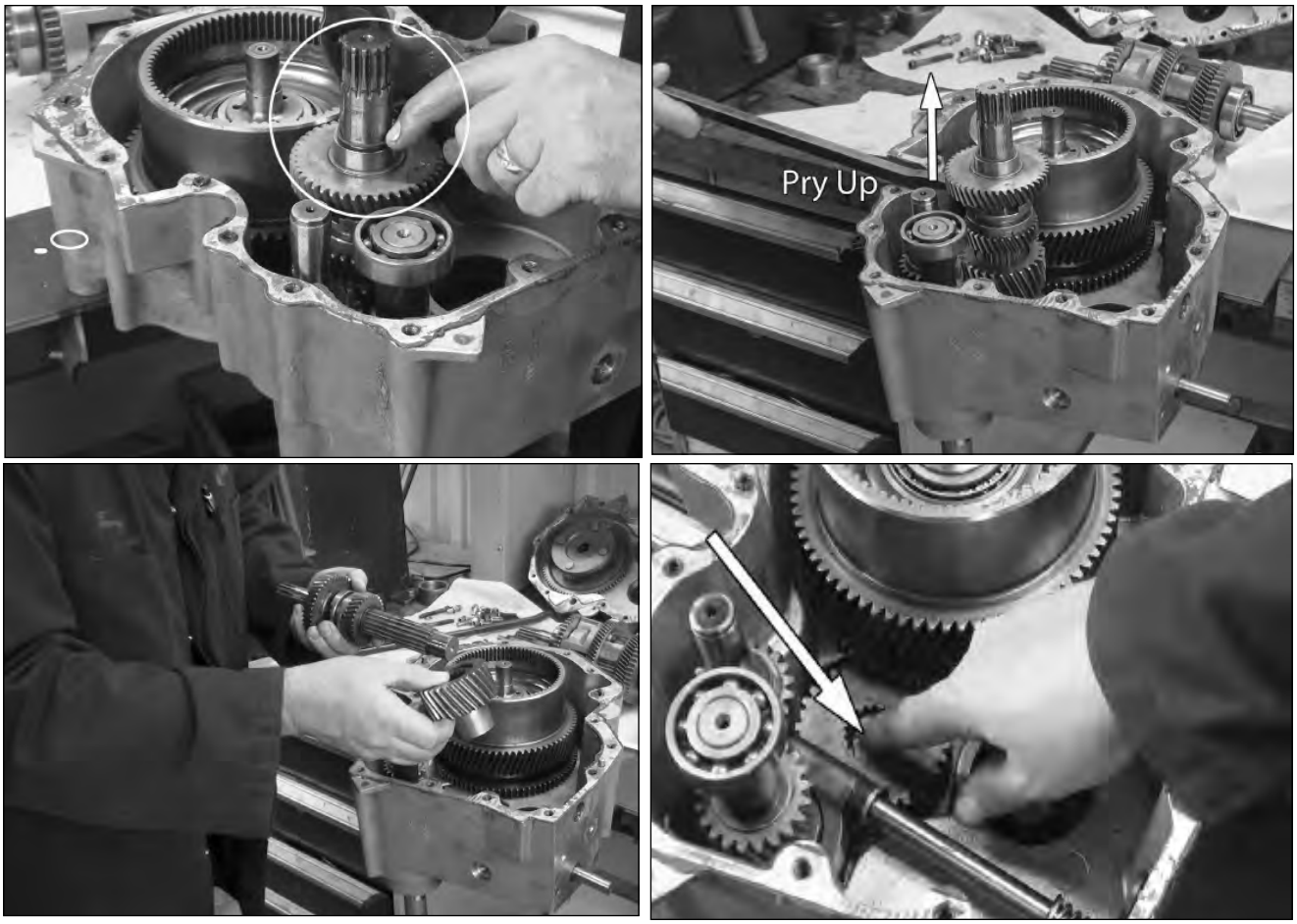
Raise the helical gear pinion from the idler shaft to allow for clearance when pulling the steering differential assembly from the housing. Move the steering differential assembly to a clean work area.



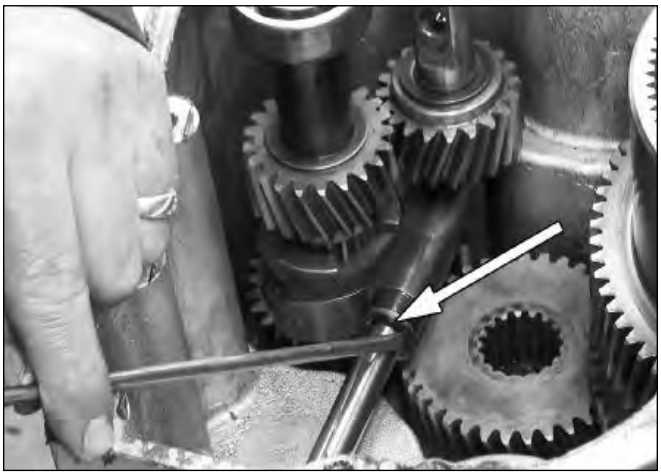
18. Temporarily place the spacer back on to the helical gear pinion and secure with the retaining ring.
19. Place a pry bar under the helical gear pinion and gently pry upwards.
20. Remove the 27 tooth helical pinion below as well as the spacer.

NOTE

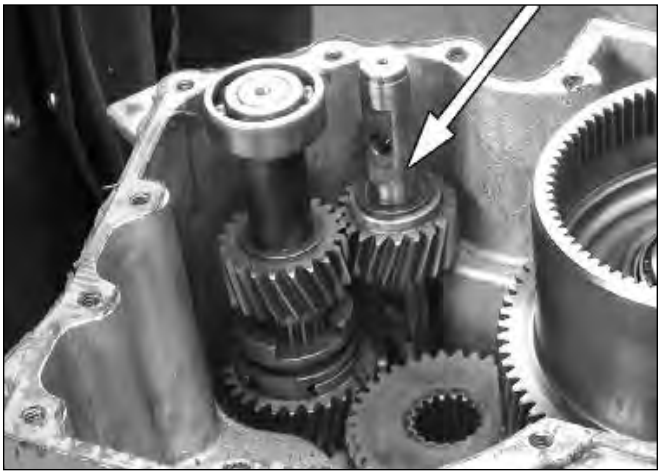
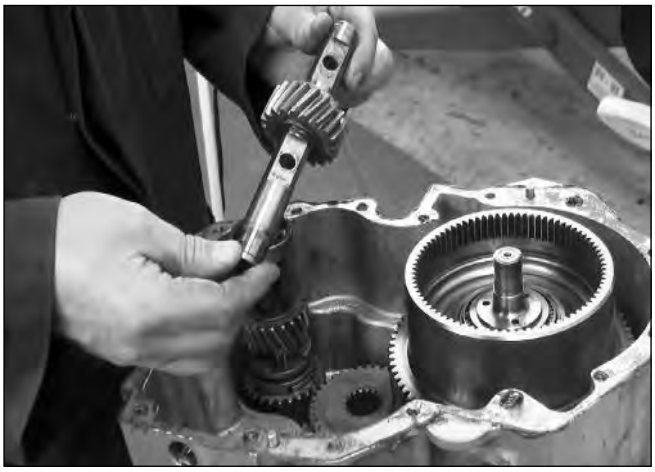
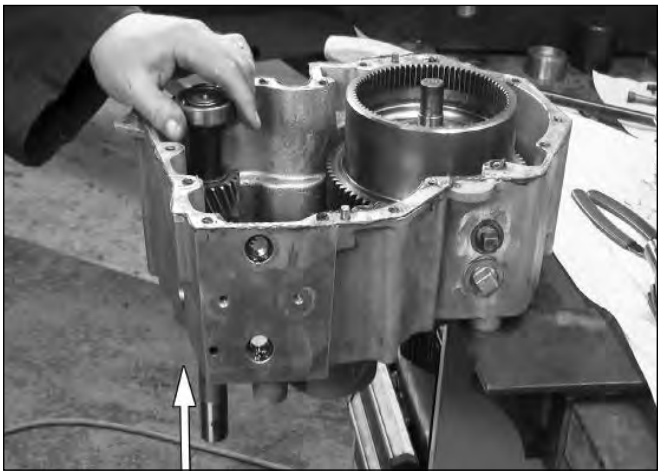
Note the remaining 35 tooth helical gear below in the housing. This will be removed at a later stage.



21. Remove the forward reverse shift shaft. Remove the retaining rings that are securing the shift shaft in the housing.

22. Gently pull the shift shaft from the housing along with the shift forks, spring, spacer and washers.
23. Pull the Reverse Idler Shaft from the housing.
24. Using a soft faced mallet, gently tap the input shaft from the housing.

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